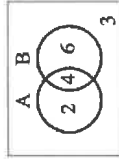


Chapter 8 Probability Worksheet B

- 1 Consider the following Venn diagram displaying the number of elements belonging to the events A and B.



Find the following probabilities:

a $\Pr(A)$ b $\Pr(A \cap B)$

c $\Pr(A | B)$ d $\Pr(B | A)$

- 2 From the 20 members of a ski club, 16 like skiing, 12 like snowboarding and 8 like both skiing and snowboarding. A ski club member is chosen at random. Let A be the event 'the person likes skiing' and B be the event 'the person likes snowboarding'.

- a Represent the information in a two-way table.

	A	A'
B		
B'		

- b Find the probability that the person only likes snowboarding.

- c Find the probability that the person likes snowboarding given that they like skiing.

- d Find the probability that the person likes skiing given that they like snowboarding.

- 3 Two four-sided dice, numbered 1 to 4, are rolled.

- a List the sample space using a table.

		Die 2			
		1	2	3	4
Die 1	1				
	2				
	3				
	4				

- b Find the probability of obtaining the outcome (3, 2).

- c Find:

i $\Pr$ (sum of at least 6)

ii $\Pr$ (sum not equal to 6)

- d Find the probability of a sum of 7, given that the sum is at least 6.

- 4 Two letters are chosen from the word NINE without replacement

- a Construct a table to list the sample space.

		Letter 2			
		N	I	N	E
Letter 1	N				
	I				
	N				
	E				



- b** Find the probability of:
- i obtaining the outcome (N, E) _____
 - ii selecting a N and a E _____
 - iii selecting two Ns _____
 - iv selecting two Ns given that at least one N is selected _____

5 Boxes A and B contain 4 counters each. Box A contains 2 red and 2 yellow counters and box B contains 3 red and 1 yellow counters. A box is chosen at random and then a single counter is selected.

- a** Find the probability of selecting a red counter from:

- i box A _____
- ii box B _____

- b** Represent the options available as a tree diagram that shows all possible outcomes and related probabilities.

- c** What is the probability of selecting box B and a red counter?

- d** What is the probability of selecting a red counter?



6 A bag contains 3 red (R) and 4 white (W) marbles and two marbles are selected without replacement.

- a** Draw a tree diagram showing all outcomes and probabilities.

- b** Find the probability of selecting:

- i a red marble and then a white marble _____
- ii 2 red marbles _____
- iii exactly 1 red marble _____

- c** If the experiment was repeated with replacement, find the answers to each question in part **b**.

- i a red marble and then a white marble _____
- ii 2 red marbles _____
- iii exactly 1 red marble _____



4 From a class of 30 students, 17 like soccer, 14 like AFL and 9 like both soccer and AFL.

a Illustrate this information in a Venn diagram.

b State the number of students who like:

i AFL only

ii neither soccer nor AFL

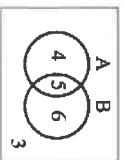
c Find the probability that a person chosen at random will like:

i AFL

ii AFL only

iii both soccer and AFL

5 This Venn diagram shows the distribution of elements in sets A and B.



a Transfer the information in the Venn diagram into a two-way table.

	A	A'	
B			
B'			

b Find:

i $n(A \cap B)$

ii $n(A' \cap B)$

iii $n(A \cap B')$

iv $n(A' \cap B')$

v $n(A)$

vi $n(B)$

vii $n(A \cup B)$

c Find:

i $\Pr(A \cap B)$

ii $\Pr(A')$

iii $\Pr(A \cap B')$

6 A card is selected from a pack of 52 playing cards. Let A be the event 'the card is a heart' and B be the event 'the card is a queen'.

a Find:

i $n(A)$

ii $n(B)$

iii $n(A \cap B)$

b Find:

i $\Pr(A)$

ii $\Pr(A')$

iii $\Pr(A \cap B)$

The sine rule, cosine rule and area of a triangle formula are usually applied to triangles that are not right-angled. For any triangle ABC:

The sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\text{or } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

The cosine rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{or } a^2 = b^2 + c^2 - 2bc \cos A$$

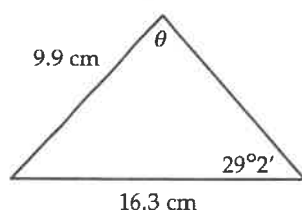
Area of a triangle

$$A = \frac{1}{2}ab \sin C$$

When using the **sine** rule, take care when finding the size of an obtuse angle as most calculators only give acute angle answers. It helps to draw a sketch of the triangle to see if there is an obtuse angle (one side much longer than the other sides is a good indication).

When using the **cosine** rule, if an angle is obtuse the calculator will show a negative cosine ratio.

e.g. 1 Find the size of θ .



Note: From the sketch θ is obtuse

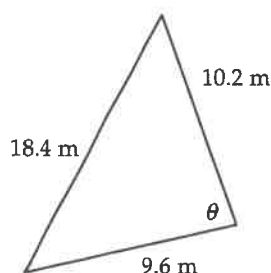
$$\frac{\sin \theta}{16.3} = \frac{\sin 29^\circ 2'}{9.9}$$

$$\sin \theta = \frac{16.3 \times \sin 29^\circ 2'}{9.9}$$

$$\therefore \theta = 53^\circ 2'$$

The required angle is obtuse, so the correct value of θ is: $180^\circ - 53^\circ 2' = 126^\circ 58'$

2 Find θ to the nearest minute.



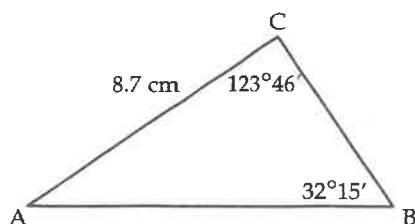
$$\cos \theta = \frac{10.2^2 + 9.6^2 - 18.4^2}{2 \times 10.2 \times 9.6}$$

$$\cos \theta = -0.7268\dots$$

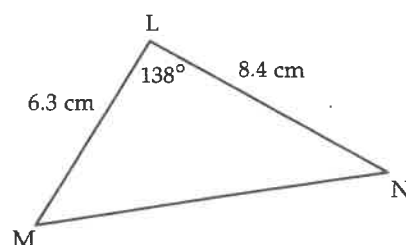
$$\therefore \theta = 136^\circ 38'$$

In the following questions, determine angles to the nearest minute, lengths to one decimal place and areas to one decimal place.

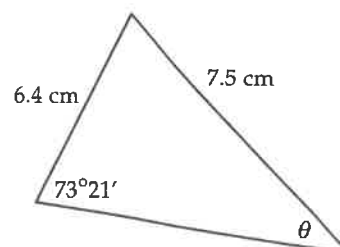
1 Determine the length AB.



2 Find the area.

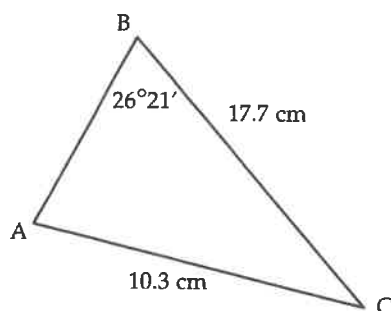


3 Determine the value of θ .

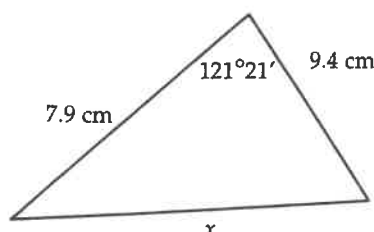


In the following questions, determine angles to the nearest minute, lengths to one decimal place and areas to one decimal place.

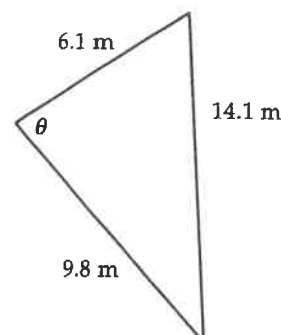
- 4 Find the size of angle A.



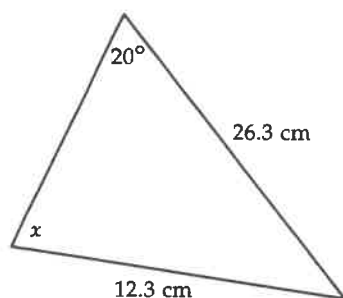
- 5 Determine the length of x .



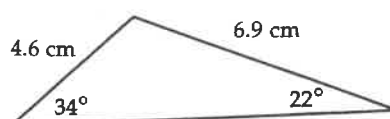
- 6 Find the value of θ .



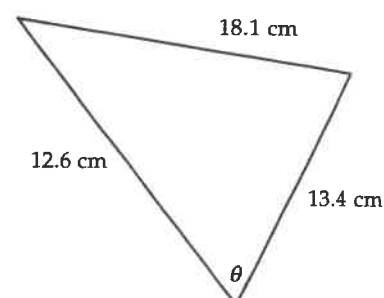
- 7 Find the value of x .



- 8 Find the area.



- 9 Find the value of θ .

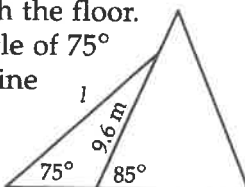


- 10 In ΔPQR , $p = 17.3$ cm, $q = 10.7$ cm and $r = 8.9$ cm. Determine the size of $\angle QPR$.

- 11 In ΔLMN , $l = 29.4$ cm, $n = 16.3$ cm and $\angle N = 29^\circ$. Determine the value of $\angle LMN$.

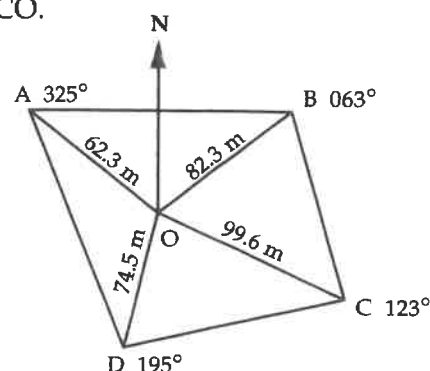
- 12 In ΔXYZ , $x = 27.3$ cm, $z = 16.8$ cm and $\angle XYZ = 23^\circ 4'$. Determine the size of the largest angle.

- 13 A builder leans a ladder against the side of an A-framed house. The sides of the house make an angle of 85° with the floor. The ladder makes an angle of 75° with the ground. Determine the length of the ladder needed to reach 9.6 m up the side of the house.

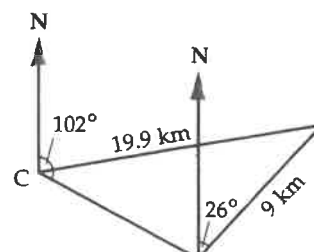


- 14 Two trawlers are in a direct line from a lighthouse. From the top of the lighthouse, the angles of depression of the two trawlers are $25^\circ 8'$ and $5^\circ 23'$ respectively. The trawlers are 350 m apart. Determine the distance from the lighthouse to the nearest trawler and the height of the lighthouse.

- 15 This diagram represents a plane table survey. It shows the bearings of points A, B, C and D. Determine the area bounded by ABCO.



- 16 A ship leaves Cairns sailing on a bearing of 102° . After some time it changes course and sails on a bearing of 026° . It travels 9 km and stops when it is 19.9 km from Cairns. Find the new bearing of the ship from Cairns.



1 13.6 cm 2 17.7 cm 3 54°50' 4 130°18' 5 15.1 cm
6 123°15' 7 133°0' 8 13.2 cm 9 88°11' 10 123°40'
11 90°1' 12 127°36' 13 9.9 m 14 88.0 m, 41.3 m
15 6088.1 cm² 16 75°58'

(acute and obtuse angles)

9.5 Sketching parabolas

Navigator

Q1 Column 1, Q2 Column 1, Q3 Column 1, Q4 Column 1, Q5 Column 1, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q16, Q17

Q1 Column 2, Q2 Column 2, Q3 Column 2, Q4 Column 2, Q5 Column 2, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q14, Q16, Q17

Q1 Column 3, Q2 Column 3, Q3 Column 3, Q4 Column 3, Q5 Column 3, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q15, Q17, Q18

Answers
page 786

Fluency

- 1 Sketch the graph of each of the following quadratic relationships by finding (i) the shape of the graph, (ii) the y -intercept, (iii) the x -intercepts, (iv) the turning point and (v) the axis of symmetry.

(a) $y = x^2 - 8x + 15$	(b) $y = x^2 - 6x + 8$	(c) $y = x^2 - 10x + 21$
(d) $y = -x^2 + 4x + 12$	(e) $y = -x^2 + 2x + 3$	(f) $y = -x^2 + 2x + 8$
(g) $y = x^2 - 8x + 12$	(h) $y = x^2 + 2x - 3$	(i) $y = x^2 + 4x - 5$

- 2 Sketch the following parabolas showing the x - and y -intercepts, the turning point and the axis of symmetry. Find the coordinates of the turning point by transforming the equations into turning point form.

(a) $y = x^2 + 2x - 8$	(b) $y = x^2 + 8x + 15$	(c) $y = x^2 + 3x + 2$
(d) $y = x^2 - 4$	(e) $y = 9 - x^2$	(f) $y = x^2 - 7$
(g) $y = x^2 - 2x$	(h) $y = x^2 + 3x$	(i) $y = x^2 + 2x$

- 3 Sketch the graphs of the following equations showing all the key features.

(a) $y = x^2 - 25$	(b) $y = x^2 - 9$	(c) $y = x^2 - 16$
(d) $y = 81 - x^2$	(e) $y = 4 - x^2$	(f) $y = 1 - x^2$
(g) $y = x^2 - 4x$	(h) $y = x^2 + 6x$	(i) $y = 3x^2 + 6x$

- 4 Sketch the graphs of the following equations showing all the key features.

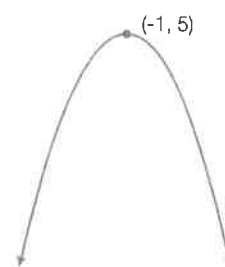
(a) $y = (x + 2)^2 - 9$	(b) $y = (x + 5)^2 - 1$	(c) $y = -(x + 1)^2 + 5$
(d) $y = (x - 3)^2 + 2$	(e) $y = (x - 2)^2 + 4$	(f) $y = (x - 4)^2 + 2$

- 5 Sketch the graphs of the following equations showing all the key features.

(a) $y = x^2 + 10x + 16$	(b) $y = x^2 + 8x + 7$	(c) $y = -x^2 - 6x - 5$
(d) $y = -x^2 + 4x - 3$	(e) $y = x^2 + 6x + 9$	(f) $y = x^2 + 10x + 25$
(g) $y = -x^2 + 3x$	(h) $y = -x^2 - 5x$	(i) $y = -x^2 + 7x - 12$

- 6 When expressed graphically, which of the following equations would have a shape like that of the parabola shown on the right?

A $y = x^2 + 2x - 5$	B $y = -x^2 - 2x + 4$
C $y = 1 - x^2$	D $y = 6 - 2x - x^2$



- 7 (a) The coordinates of the y -intercept of $y = x^2 + 4x + 3$ are:

A (0, 8)	B (-1, 0)	C (3, 0)	D (0, 3)
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- (b) The coordinates of the x -intercepts of $y = x^2 - 9$ are:

A (0, -3), (0, 3)	B (-3, 0), (3, 0)	C (-9, 0), (9, 0)	D (0, -9), (0, 9)
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2 Plot the graph of $y = x^2 - 2x - 8$.

- (a) Write down the coordinates of the x - and y -intercepts.
 (b) Write down the coordinates of the turning point. Does the parabola have a maximum or a minimum turning point?

(c) Write down the equation of the axis of symmetry.

(d) From the graph, find the value of y when $x = 3$.

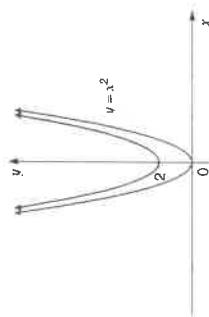
(e) What are the values of x when $y = 7$?

(f) Hence, state the values of x when $x^2 - 2x - 8 = 7$.

(g) Write down the values of x when $x^2 - 2x - 8 = 0$.

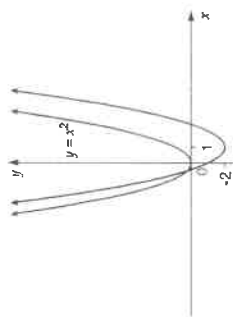
3 (a) The equation of the parabola shown could be:

- A $y = x^2 + 2$
 B $y = x^2 - 2$
 C $y = (x - 2)^2$
 D $y = (x + 2)^2$



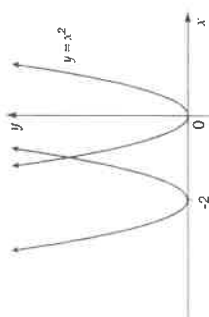
(b) The equation of the parabola shown could be:

- A $y = (x - 1)^2 - 2$
 B $y = (x - 1)^2 + 2$
 C $y = (x + 1)^2 + 2$
 D $y = (x + 1)^2 - 2$



(c) The equation of the parabola shown could be:

- A $y = (x - 2)^2$
 B $y = (x + 2)^2$
 C $y = x^2 - 2$
 D $y = x^2 + 2$



4 Solve these quadratic equations using the Null Factor Law.

- (a) $x^2 - 6x = 0$ (b) $4x^2 - 81 = 0$ (c) $x^2 - 5x - 14 = 0$
 (d) $x^2 + 6x + 9 = 0$ (e) $x^2 = 8x$ (f) $x^2 + 36 = 12x$
 (g) $2x(x + 4) = 0$ (h) $(x - 7)(x + 1) = 0$ (i) $x^2 - 16 = 0$

5 Solve these quadratic equations.

- (a) $x^2 + 12x + 35 = 0$ (b) $x^2 + x - 6 = 0$ (c) $-x^2 - 8x + 9 = 0$
 (d) $-x^2 + 9x - 8 = 0$ (e) $3x^2 + 9x - 12 = 0$ (f) $2x^2 - 8x + 6 = 0$

6 Solve these quadratic equations.

- (a) $x^2 - 4x = 5$ (b) $x^2 - 6 = 5x$ (c) $x^2 + 12 = -7x$
 (d) $3x^2 = 27$ (e) $(x - 3)(x + 7) = 11$ (f) $(x - 3)(x + 2) = 6$

Ex 9.4

- 7 Solve each of the following by completing the square.
 (a) $x^2 - 4x + 2 = 0$ (b) $x^2 + 8x + 9 = 0$ (c) $-x^2 - 2x + 4 = 0$

Ex 9.5

- 8 For each of the following:
 (i) state the changes that have been made to the graph of $y = x^2$
 (ii) state the coordinates of the turning point
 (iii) find the x - and y -intercepts
 (iv) sketch a graph of each equation showing the turning point and intercepts.
 (a) $y = x^2 - 4$ (b) $y = -2(x + 1)^2 - 4$ (c) $y = \frac{1}{3}(x - 7)^2$

Ex 9.5

- 9 For each of the following equations:
 (i) find the y -intercept
 (ii) find the x -intercepts
 (iii) find the coordinates of the turning point
 (iv) sketch the graph.
 (a) $y = x^2 - 4x - 12$ (b) $y = -(x + 1)^2 + 4$
 (c) $y = 2(x - 3)^2 - 6$ (d) $y = -x^2 + 3x + 4$
 10 Sketch the graph of each of the following.
 (a) $(x + 2)^2 + (y - 3)^2 = 16$ (b) $x^2 + (y - 3)^2 = 4$
 (c) $y = \frac{1}{x - 4} + 1$ (d) $y = 2x^2 + 2 - 4$

Ex 9.6

Understanding

11 A softball thrown by an outfielder to the pitcher followed a path where the height h , in metres, of the softball after t seconds, is described by the relationship:

$$h = -t^2 + 6t + 1$$

- (a) Complete a table of values for this relationship.
 (b) Plot the graph of this relationship on graph paper.
 (c) What was the height of the softball after 2 seconds?
 (d) When was the ball first at a height of 7 m above the ground?
 (e) During which time interval was the ball above a height of 9 m?
 (f) What was the maximum height of the softball during its flight?
 (g) How long was the ball in flight if it was caught at a height of 1.6 m above the ground on its downward path?

12 A train ticket is 5 cm longer than it is wide.

- (a) Draw a diagram of the ticket, labelling the width x .
 (b) Write down the length of the ticket in terms of x .
 (c) Write an expression for the area of the train ticket.
 (d) If each ticket requires 24 cm² of paper, write an equation that represents this area.
 (e) Solve this equation to find x .
 (f) Hence, state the dimensions of the train ticket.
 (g) Did you use both of the values for x that you found using algebra to find the dimensions of the ticket? Explain your answer.

Ex 9.3

Ex 9.3

Ex 9.3

